

Dataset	Number of Workers	Number of Items
Hashimoto et al. (2019) (LM)	124	50
Hashimoto et al. (2019) (summarization)	96	99
Hashimoto et al. (2019) (Reddit)	123	99
WMT19	176	1997
Dathathri et al. (2020)	15	1358
Holtzman et al. (2020)	140	1399

Table 12: Number of workers and items in each of our convenience sampled datasets.

Dataset	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\beta}_4$	$\hat{\beta}_5$	$\hat{\beta}_6$	$\hat{\sigma}_{wi}$
Hashimoto et al. (2019) (LM)	0.55	-0.03						0.25
Hashimoto et al. (2019) (summarization)	0.58	0.06						0.26
Hashimoto et al. (2019) (Reddit)	0.55	0.05	0.03	0.01				0.23
WMT19	0.86	0.04						0.12
Dathathri et al. (2020)	0.62	0.04	-0.05	-0.03				0.16
Holtzman et al. (2020)	0.59	0.02	0.04	0.02	0.01	0	-0.04	0.16

Table 13: Fit fixed effect coefficients for each model along with the residual model variance. If only one model is compared to a baseline, there is a value for intercept and β_1 . If more than one model, there is an additional parameter for each model. Because we use contrast coding, each coefficient can be interpreted as the difference from the grand mean.

Dataset	$\hat{\sigma}_{0w}$	$\hat{\sigma}_{1w}$	$\hat{\sigma}_{2w}$	$\hat{\sigma}_{3w}$	$\hat{\sigma}_{4w}$	$\hat{\sigma}_{5w}$	$\hat{\sigma}_{6w}$
Hashimoto et al. (2019) (LM)	0	0.11	0.11				
Hashimoto et al. (2019) (summarization)	0	0.13	0.11				
Hashimoto et al. (2019) (Reddit)	0.11	0.04	0.08	0.06	0.17		
WMT19	0.07	0.04	0.13				
Dathathri et al. (2020)	0	0.04	0.05	0.05	0.05		
Holtzman et al. (2020)	0.09	0.05	0.03	0.04	0.04	0.02	0.04

Table 14: Fit random effects standard deviations for worker. As in the equations above, $\hat{\sigma}_{0w}$ is the worker intercept and the rest of the parameters are worker slopes for each model.

Dataset	$\hat{\sigma}_{0i}$	$\hat{\sigma}_{1i}$	$\hat{\sigma}_{2i}$	$\hat{\sigma}_{3i}$	$\hat{\sigma}_{4i}$	$\hat{\sigma}_{5i}$	$\hat{\sigma}_{6i}$
Hashimoto et al. (2019) (LM)	0.04	0.14	0.1				
Hashimoto et al. (2019) (summarization)	0.07	0	0.18				
Hashimoto et al. (2019) (Reddit)	0	0.13	0.11	0.14	0.14		
WMT19	0.05	0.03	0.15				
Dathathri et al. (2020)	0	0.16	0.19	0.16	0.16		
Holtzman et al. (2020)	0	0.13	0.1	0.12	0.11	0.13	0.13

Table 15: Fit random effects standard deviations for item. As in the equations above, $\hat{\sigma}_{0i}$ is the item intercept and the rest of the parameters are item slopes for each model.

Scenario	σ_{w0}	σ_{w1}	σ_{i0}	σ_{i1}	σ_{wi}
Low variance	0.01	0.04	0.01	0.13	0.16
High variance	0.01	0.11	0.04	0.14	0.26

Table 16: An example of high variance and low variance settings. The standard deviations correspond to the variance parameters for worker intercept, worker slope, item intercept, item slope, and sigma, respectively.

I Additional Plots of Model Overlap



